

Invited Paper

Toward the development of a Trust-Tech-based methodology for solving mixed integer nonlinear optimization

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Abstract: Many applications of smart grid can be formulated as constrained optimization problems. Because of the discrete controls involved in power systems, these problems are essentially mixed-integer nonlinear programs. In this paper, we review the Trust-Tech-based methodology for solving mixed-integer nonlinear optimization. Specifically, we have developed a two-stage Trust-Tech-based methodology to systematically compute all the local optimal solutions for constrained mixed-integer nonlinear programming (MINLP) problems. In the first stage, for a given MINLP problem this methodology starts with the construction of a new, continuous, unconstrained problem through relaxation and the penalty function method. A corresponding dynamical system is then constructed to search for a set of local optimal solutions for the unconstrained problem. In the second stage, a reduced constrained NLP is defined for each local optimal solution by determining and fixing the values of integral variables of the MINLP problem. The Trust-Tech-based method is used to compute a set of local optimal solutions for these reduced NLP problems, from which the optimal solution of the original MINLP problem is determined. A numerical simulation of several testing problems is provided to illustrate the effectiveness of our proposed method.

Key Words: mixed integer nonlinear programming, Trust-Tech methodology, global optimization, dynamical system methods

1. Introduction

A smart grid needs to be an optimized power grid for efficiency and reliability. Optimized operational efficiency, whether in an economic or physical sense, is one of the most important attributes of a smart grid. Optimization technology has a long history of being applied to solve many power system operation and planning problems. Many applications, such as power system planning [1–3], unit commitment problems [4–6], placement of synchronized measurements [7] and capacitor placement and control [8–11], fall under the category of nonlinear constrained optimization problems (NLP). On the other hand, because of the discrete controls (shunt compensation, transformer taps, etc) involved in power systems, these optimization problems are essentially mixed-integer nonlinear programs.

Mixed integer nonlinear programming (MINLP) provides a fairly general and powerful framework

to model optimization problems involving both discrete and continuous variables. In this framework, the nonlinear expression better approximates real-world phenomena while its discrete variables offer great flexibility to represent an indivisible quantity or applications involving decision making. In the literature, numerous studies of MINLP problems have been reported in areas such as engineering [12, 13], biological science and economics. For instance, determination of the molecular structure of a substance based on the energy calculation is formulated as an MINLP problem [14–16]. The MINLP methodology was applied to portfolio selection in financial engineering [17, 18] and to resource allocation and scheduling problems [19, 20]. Over the past few decades, there has been a pronounced increase in the study of MINLP problems.

Although MINLP provides great flexibility and accuracy, the task of solving such problems is usually challenging. The diverse nonlinear behavior in MINLP problems creates a great challenge not only in algorithmic developments but also in numerical implementations. Developing efficient and robust methods and tools for solving MINLP problems continues to attract a lot of research and development efforts. From a theoretical viewpoint, it is well known that the major difficulty in solving general continuous nonlinear programming problems arises from the non-convexity of objective functions and/or nonlinear constraints. Without the aid of other supplemental techniques, the performance of stand-alone, deterministic, local-type algorithms is greatly restricted by such non-convexity. The presence of integral variables further deteriorates the situation since it leads to an explosive increase in the number of solutions.

A large variety of algorithms have been proposed for solving MINLP problems. Many of these proposed algorithms basically integrate and extend schemes designed for handling non-convexity arising from nonlinear expressions and the presence of integral variables. Algorithms of this type reported in the literature include Branch and Bound [21, 22], Generalized Benders Decomposition [23], Outer Approximation [24], Extended Cutting Plane Method [25] and so forth. Details of these proposed algorithms may be found in [26].

Although schemes such as the branching or cutting plane method might alleviate adverse effects due to the presence of discrete variables, these algorithms are still limited by the non-convex nature arising from the inherent nonlinear expression. For instance, the Branch and Bound algorithm creates a sequence of sub-problems from the original MINLP problem and seeks to identify optimal solutions by solving some sub-problems. Its overall efficiency depends heavily on whether good solutions may be identified at an early stage to help its pruning scheme. To identify potential solutions, it will be indispensable to solve some non-convex sub-problems (even without integral variables).

The Trust-Tech methodology has been developed to solve non-convex global optimization problems [28–30] and it has been successfully applied to various applications, such as learning finite mixture models [31], constructing neural network ensembles [32], and solving optimal power flow problems [33]. One distinguishing feature of the Trust-Tech methodology comes from its ability to systematically and efficiently locate multiple local optimal solutions in a tier-by-tier manner. With the aid of topological information from the problem structure, the issue of non-convexity can be successfully overcome. We have demonstrated its efficiency in general continuous nonlinear programming problems. In this paper we present a Trust-Tech-based methodology to conquer the non-convexity problem both from nonlinear expression and the presence of integral variables. However, we point out that the Trust-Tech methodology presented for solving MINLP problems is by no means the only possible extension of the Trust-Tech methodology. There are several possible ways that the Trust-Tech methodology can significantly contribute to solving MINLP problems. For instance, the Trust-Tech methodology can obviously be incorporated with Branch and Bound type algorithms to enhance their effectiveness.

The rest of this paper is organized as follows. An overview of the Trust-Tech methodology for general continuous optimization is first given in Section 2. In Section 3, we present the proposed Trust-Tech based methodology in detail for solving mixed-integer nonlinear programs. We then give numerical simulations on illustrative examples in Section 4 and conclude the paper with remarks in Section 5.

2. Overview of Trust-Tech methodology for continuous optimization

We consider a general constrained nonlinear optimization problem of the form:

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in I = \{1, \dots, l\}, \\ & g_j(x) \leq 0, \quad j \in J = \{1, \dots, s\}. \end{aligned} \quad (1)$$

The constraint set of (1) defines the following feasible set:

$$S = \{x \in R^n : h_i(x) = 0, i \in I, \quad g_j(x) \leq 0, j \in J\},$$

which can be any closed subset of R^n , and its structure can be very complex. The set S is usually non-convex and disconnected; i.e. it is composed of several disjoint and path-connected feasible regions. The task of locating each connected feasible region of the set S is in itself a difficult one.

Without loss of generality, we consider the following optimization problem with equality constraints:

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, i \in I = \{1, \dots, m\}. \end{aligned} \quad (2)$$

We define the following equality vector $H(x) := (h_1(x), \dots, h_m(x))^T$. By applying the regularity condition and Sard's theorem [36], it can be shown that the constraint set (or the feasible set or region) defined by

$$M = \{x \in R^n : H(x) := (h_1(x), \dots, h_m(x))^T = 0\},$$

is a smooth manifold. In general, the extremely complicated constraint set M can be decomposed into several disjoint path-connected feasible components.

One challenge in solving the nonlinear optimization problem (2) is that the feasible set M can be very complex and contain multiple feasible components. The great majority of existing numerical methods/techniques and heuristics for solving constrained nonlinear optimization problems are usually globally convergent and can only verify the existence of feasible solutions after a feasible solution is found. In other words, when these methods fail to acquire a feasible solution, they usually can not verify whether such infeasibility is local or global.

2.1 Mathematical preliminary

We first consider a nonlinear dynamical system described by

$$\dot{x} = F(x), \quad (3)$$

where $F : M \rightarrow TM$ is C^1 and M is a manifold in R^n . Here $F(x) \in T_x M$ is a tangent vector to M at x . A state vector $\hat{x}$ is called an *equilibrium point (EP)* of system (3) if $F(\hat{x}) = 0$. We say that an equilibrium point of (3) is *hyperbolic* if the Jacobian of $F(\cdot)$ at $\hat{x}$, denoted $J_F(\hat{x})$, has no eigenvalue with a zero real part. If the Jacobian of the hyperbolic equilibrium point has exactly k eigenvalues with positive real parts, we call it a *type- k equilibrium point*. It can be shown that for a hyperbolic equilibrium point, it is a (asymptotically) *stable equilibrium point (SEP)* if all the eigenvalues of its corresponding Jacobian have negative real parts; otherwise it is an *unstable equilibrium point (UEP)*.

The solution curve of (3) starting from x at $t = 0$ is called the system *trajectory* starting from x and is denoted by $\Phi(t, x)$. For a type- k hyperbolic EP x^* , its stable and unstable manifolds $W^s(x^*)$ and $W^u(x^*)$ are defined as follows:

$$\begin{aligned} W^s(x^*) &= \{x \in M : \lim_{t \rightarrow +\infty} \Phi(t, x) = x^*\}, \\ W^u(x^*) &= \{x \in M : \lim_{t \rightarrow -\infty} \Phi(t, x) = x^*\}, \end{aligned} \quad (4)$$

where the dimensions of $W^u(x^*)$ and $W^s(x^*)$ are k and $n - k$, respectively. A dynamical system is called completely stable if every trajectory of the system approaches one of its EPs. The stability region (or region of attraction) of an SEP x_s is defined as:

$$A(x_s) := \{x \in M : \lim_{t \rightarrow +\infty} \Phi(t, x) = x_s\}. \quad (5)$$

The quasi-stability region $A_p(x_s)$ of an SEP x_s is defined as [18]:

$$A_p(x_s) = \text{int}(\overline{A(x_s)}). \quad (6)$$

From a topological point of view, $A(x_s)$ and $A_p(x_s)$ are open, invariant and connected sets. A type-1 EP lying on the quasi-stability boundary $\partial A_p(x_s)$ of an SEP x_s is called a *decomposition point* of x_s . Note that a type-1 EP lying on the stability boundary may not be a decomposition point while a decomposition point lies not only on the quasi-stability boundary but also on the stability boundary.

Another system encountered in Trust-Tech methodology is the following nonlinear non-hyperbolic dynamical system described by

$$\dot{x} = F(x) = M(x)H(x), \quad (7)$$

where $H : R^n \rightarrow R^m$ and $M : R^n \rightarrow R^{n \times m}$, $n \geq m$, are all C^r , $r \geq 1$ for the existence and uniqueness of solutions to system (7). System (7) is called non-hyperbolic because its equilibria are no longer isolated. Such a non-hyperbolic phenomenon is unavoidable because the zero set $E_F \equiv F^{-1}(0) = M^{-1}(0) \cup H^{-1}(0)$, which is a manifold, usually consists of several path-connected components. A path-connected (compact) component of the zero set $F^{-1}(0)$, say Σ , is called an *equilibrium manifold* of system (7). An l -dimensional equilibrium manifold Σ of (7) is called *pseudo-hyperbolic* or just *hyperbolic* if for each $x \in \Sigma$, the Jacobian of $F(\cdot)$ at x , denoted by $J_F(x)$, has no eigenvalues with zero real parts on the normal space $N_x(\Sigma)$, the orthogonal complement of the tangent space $T_x(\Sigma)$, of Σ at x in R^n and there exists an $\epsilon > 0$ such that $\Phi_{-\infty} : B_\epsilon(\Sigma) \rightarrow \Sigma$, defined by $B_\epsilon(\Sigma) = \{x \in R^n : \|x - y\| < \epsilon, \forall y \in \Sigma\}$ and $\Phi_{-\infty}(x) = \lim_{t \rightarrow -\infty} \Phi(t, x)$, is locally homeomorphic to a projection from R^n to R^l . A hyperbolic equilibrium manifold Σ is a (asymptotically) *stable equilibrium manifold* if for each $x \in \Sigma$, all the eigenvalues of its corresponding Jacobian have a negative real part; otherwise, it is an *unstable equilibrium manifold*. For a hyperbolic equilibrium manifold Σ , its stable and unstable manifolds $W_F^s(\Sigma)$ and $W_F^u(\Sigma)$ are defined similarly [34]. Furthermore, for a stable equilibrium manifold Σ_s , its stability region $A(\Sigma_s)$ and stability boundary $\partial A(\Sigma_s)$ can also be defined.

2.2 Trust-Tech methodology

We believe that the only reliable way to find the global optimal solution of the optimization problem (2) is to first find all the local optimal solutions and then find, from them, the global optimal solution. To this end, we have developed the *Transformation under stability-retained equilibrium characterization* (Trust-Tech) methodology to systematically compute all the local optimal solutions of (2) via the following two phases:

Phase I: Find all the feasible components of the constraint (feasible) set based on the trajectories of one dynamical system.

Phase II: Find all the local optimal solutions in each feasible component of the constraint set based on the trajectories of another dynamical system.

2.2.1 Phase I

Phase I finds all the feasible components via some trajectories of a nonlinear dynamical system. In order to visit each path-connected feasible component of the constraint set M , Phase I consists of two steps:

Step 1.1: Approach a path-connected feasible component of the optimization problem (2).

Step 1.2: Escape from the path-connected feasible component and approach another path-connected feasible component of the optimization problem (2).

We design a nonlinear dynamical system whose trajectories can be used to perform Step 1.1 and Step 1.2. The central idea in designing the nonlinear dynamical system is that every path-connected feasible

component M_k corresponds to a stable equilibrium manifold of the nonlinear dynamical system. In this way, the task of finding all the path-connected feasible components of the optimization problem (2) can be accomplished via finding all the stable equilibrium manifolds of the nonlinear dynamical system. One way to achieve this goal (or to implement the above idea) is to build the following so-called quotient gradient system (QGS) associated with the constraint set characterized by $H(x)$:

$$\dot{x} = -DH(x)^T H(x), \quad (8)$$

where $DH(x)$ represents the Jacobian matrix of the vector $H(x)$.

It can be shown that each feasible component of the optimization problem (2) can be located via each stable equilibrium manifold of the QGS (8) [34]. In addition, the stability boundary of a stable equilibrium manifold can be characterized as the union of the stable manifolds of all equilibrium manifolds on the stability boundary [34]. These analytical results provide a basis for the Trust-Tech-based methodology when searching for feasible components. To numerically implement Phase I of the methodology, we propose the following two tasks:

Task 1.1: Approach a stable equilibrium manifold of the QGS (8).

Task 1.2: Escape from the stable equilibrium manifold and approach another stable equilibrium manifold of the QGS (8).

Task 1.1 can be numerically implemented by following the trajectory of the QGS (8) starting from any initial point (which can be an infeasible point of the optimization problem (2)) to a point located in the stable manifold of system (8) (i.e. a point located in a feasible component of the optimization problem (2)). Task 1.2 can be numerically implemented by following the trajectory in reverse time starting from an initial point in the stable manifold of the QGS (8), until the trajectory approaches a point in an unstable equilibrium manifold on the stability boundary of the stable equilibrium manifold. Then by following the trajectory starting from an initial point, which is close to the unstable equilibrium manifold but outside the stability region, we can approach another stable equilibrium manifold of the QGS (8), which is another feasible component of the optimization problem (2).

2.2.2 Phase II

Phase II finds all the local optimal solutions in each feasible component based on the trajectories of another nonlinear dynamical system. In order to visit all the local optimal solutions located in each feasible component, Phase II is conceptually composed of two steps:

Step 2.1: Starting from a point in the feasible component found in Phase I, approach a local optimal solution located in the feasible component of the optimization problem (2).

Step 2.2: Escape from the local optimal solution and approach another local optimal solution of the same feasible component of the optimization problem (2).

We propose to design a nonlinear dynamical system whose trajectories can be used to perform Step 2.1 and Step 2.2. The central idea in designing the nonlinear dynamical system is that all the local optimal solutions of the optimization problem (2) correspond to all the SEPs of the nonlinear dynamical system; in particular, every local optimal solution of the optimization problem (2) corresponds to an SEP of the nonlinear dynamical system. To this end, we design the following projected gradient dynamical system (PGS):

$$\dot{x} = -P_H(x(t))\nabla f(x(t)), \quad (9)$$

where $x(0) = x_0 \in M$ and the projection matrix

$$P_H(x) = (I - DH(x)^T(DH(x)DH(x)^T)^{-1}DH(x)) \in R^{n \times n},$$

is a positive semi-definite matrix for every $x \in M$. Note that $P_H(x(t))\nabla f(x(t))$ is the orthogonal projection of $\nabla f(x)$ to the tangent space $T_x M$, which means $P_H(x)\nabla f(x) \in T_x M$ for all $x \in M$.

Hence, every trajectory of (9) starting from $x_0 \in M_k$ stays in M_k . In other words, M_k is an invariant set of the PGS (9).

It can be shown that a local optimal solution of the optimization problem (2) can be located via each SEP of the PGS (9) [29]. In addition, the stability boundary of an SEP can be characterized as the union of the stable manifolds of all EPs on the stability boundary [35]. These analytical results provide a basis for the Trust-Tech-based methodology in searching for local optimal solutions. To numerically implement Phase II of the methodology, we propose that the following two tasks be numerically implemented:

Task 2.1: Approach a stable equilibrium point of the PGS (9).

Task 2.2: Escape from the stable equilibrium point and approach another stable equilibrium point of the PGS (9).

Task 2.1 can be numerically implemented by following the trajectory of the PGS (9) starting from a point in the feasible component, which is computed in Phase I, to an SEP also located in the feasible component, which is a local optimal solution to the optimization problem (2). Task 2.2 can be numerically implemented by following the trajectory in reverse time starting from an initial point in the stability region of the PGS (9), until the trajectory approaches a UEP on the stability boundary of the SEP. Then by following the trajectory starting from an initial point, which is close to the UEP but outside the stability region, we can approach another SEP of the PGS (9), which another local optimal solution to the optimization problem (2).

2.3 Theoretical basis

We present in this section a theoretical basis of the two-phase dynamical trajectory-based methodology. It has been shown that certain trajectories of the QGS (8) connects all the disjoint feasible components of the constrained optimization problem (2) [34]. These trajectories are used in Phase I to find all the disjoint feasible components. It has been also shown that certain trajectories of PGS (9) connects all the local optimal solutions located in each feasible component of the optimization problem (2) [29]. These dynamical trajectories are used in Phase II to find all the local optimal solutions located in each feasible component of the optimization problem (2). The following theoretical analysis is based on the assumption that the optimization problem (2) is regular.

Theorem 2.1 (Completely stable). *If the optimization problem (2) is regular, then the corresponding PGS (9) is completely stable and the objective function $f(x)$ of the optimization problem (2) is bounded from below on the constraint set M .*

Theorem 2.2 (Characterization of quasi-stability boundary [35]). *Let $A_p(x_s)$ be the quasi-stability region of x_s of the PGS (9) and let $x_i, i = 1, 2, \dots$ be all the decomposition points of x_s . If the system satisfies the following conditions:*

- i) The EPs on $\partial A_p(x_s)$ are hyperbolic and finite in number*
- ii) The stable and unstable manifolds of the EPs on $\partial A_p(x_s)$ satisfy the transversality condition*

then the quasi-stability boundary $\partial A_p(x_s)$ is the union of the closure of the stable manifold of x_i ; in other words

$$\partial A_p(x_s) = \bigcup_{i=1,2,\dots} \overline{W^s(x_i)}. \quad (10)$$

The above two theorems lead to the following conclusions: (i) the state space of the PGS (9) is composed of the closure of the quasi-stability region of each SEP; (ii) the boundary of the quasi-stability region (quasi-stability boundary) is the union of the closure of the stable manifolds of all the decomposition points on the quasi-stability boundary; and (iii) every trajectory of the PGS (9) converges, with probability-one, to one of SEPs.

We next explore the relationship between the local optimal solutions of the optimization problem and the EPs of the PGS (9).

Theorem 2.3 (EPs and local optimal solutions). *If $\bar{x}$ is a non-degenerate critical point, then*

- i) $\bar{x}$ is a hyperbolic EP of the PGS (9)*
- ii) $\bar{x}$ is an SEP of (9) if and only if $\bar{x}$ is an isolated local minimum of (2)*
- iii) $\bar{x}$ is a source of (9) if and only if $\bar{x}$ is an isolated local maximum of (2).*

Theorem 2.4 (Decomposition points and the SEPs). *Let $A_p(x_s)$ be the quasi-stability region of x_s of the PGS (9) satisfying the following conditions: (i) the EPs on the quasi-stability boundaries are hyperbolic and finite in number; (ii) the stable and unstable manifolds of the EPs on the quasi-stability boundaries satisfy the transversality condition. If x_d is a decomposition point on the quasi-stability boundary $\partial A_p(x_s)$, then there exists another unique SEP, denoted as $\bar{x}_s$, to which the unstable manifold of x_d converges. Conversely, if the set $\{\partial A_p(x_s) \cap \partial A_p(\bar{x}_s)\}$ is non-empty, then the set contains a decomposition point.*

Theorem 2.3 in conjunction with Theorem 2.4 suggests a conceptual computational scheme to escape from a local optimal solution and to move towards another local optimal solution via the following two steps:

Step 1: Start from a local optimal solution to the optimization problem (2), which is an SEP, denoted as x_s , of the PGS (9) (by Theorem 2.3), and move in reverse time along the unstable manifold of a decomposition point, denoted as x_d , on the quasi-stability boundary $\partial A_p(x_s)$ until x_d is reached;

Step 2: Find another local optimal solution by starting from a point near x_d but outside the $\partial A_p(x_s)$ and moving in forward time along the unstable manifold of x_d until another SEP of the PGS (9) is reached.

Similar theoretical results are available for the QGS (8).

Theorem 2.5 (Completely stable). *Suppose that the QGS (8) satisfies the following conditions: (i) all the equilibrium manifolds are hyperbolic and finite in number; and (ii) either $\|H(x)\|$ is a proper map or for each $x_0 \in R^n$, we have $\inf\{\|Q_H(x)\| : x \in K_{x_0}\} > 0$ for any closed subset K_{x_0} of the set $\{x \in R^n : \|H(x)\| \leq \|H(x_0)\|, Q_H(x) \neq 0\}$. Then the QGS (8) is completely stable.*

Theorem 2.6 (Characterization of the stability boundary). *Let Σ_s be a stable equilibrium manifold of the QGS (8) and suppose the following assumptions are held:*

- i) If an equilibrium manifold Σ has a nonempty intersection with $\partial A(\Sigma_s)$, then $\Sigma \subset \partial A(\Sigma_s)$.*
- ii) All the equilibrium manifolds on $\partial A(\Sigma_s)$ are hyperbolic and have the same dimensions.*
- iii) The stable and unstable manifolds of equilibrium manifolds on $\partial A(\Sigma_s)$ satisfy the transversality condition.*

Let $\Sigma_i, i = 1, 2, \dots$ be all the equilibrium manifolds on $\partial A(\Sigma_s)$, then

$$\partial A(\Sigma_s) = \bigcup_{\Sigma_i \in \partial A(\Sigma_s)} W^s(\Sigma_i). \quad (11)$$

Similarly, the above two theorems lead to the following conclusions: (i) the state space of the QGS (8) is composed of the closure of the stability region of each stable equilibrium manifold; (ii) the boundary of the stability region (stability boundary) is the union of the stable manifolds of all the equilibrium manifolds on the stability boundary; and (iii) every trajectory of the QGS (8) converges, with probability-one, to one of the stable equilibrium manifolds.

Theorem 2.7 (Equilibrium manifolds and feasible regions [29]). *Assume that all the equilibrium manifolds of the QGS (8) are hyperbolic and finite in number. Then, (i) each connected component of the constraint set M is a stable equilibrium manifold of (8), and (ii) if Σ is a stable equilibrium manifold of (8), then Σ is a local optimal solution to the minimization problem*

$$\min_{x \in R^n} E(x) = \frac{1}{2} \|H(x)\|^2. \quad (12)$$

Theorem 2.7 suggests a conceptual computational scheme to escape from a feasible component and to move towards another feasible component via the following steps:

Step 1: Start from a local optimal solution to the optimization problem (12), which is in a stable equilibrium manifold (denoted as Σ_s) of the QGS (8), and move in reverse time along the unstable manifold of an unstable equilibrium manifold (denoted as Σ_u) on the stability boundary $\partial A(\Sigma_s)$ until Σ_u is reached;

Step 2: Find another local optimal solution to the optimization problem (12) by starting from a point near the unstable equilibrium manifold but outside $\partial A(\Sigma_s)$ and moving in forward time along the unstable manifold of Σ_u until another equilibrium manifold of the QGS (8) is reached.

It should be noted that only those stable equilibrium manifolds of the QGS (8) having a function value $E(x) = 0$ will be the actual feasible components of the optimization problem (2).

2.4 Comparison with other trajectory methods

The idea of using dynamical system concepts to solve global optimization problems is not new. A good review is provided by Diener [38] for the application of trajectory methods to unconstrained global optimization problems. For equality constrained optimization problems, Yamashita [37] defined a nonlinear autonomous system and developed a trajectory continuation method to find multiple critical points in the system, which are revealed to be local optimal solutions to the optimization problem. This method, however, is only capable of finding multiple local optimal solutions in a single bounded and connected feasible region. In [39], Yang *et al.* combined two trajectory methods to find multiple stationary points of an objective function. Their first method used a homotopy scheme to trace a trajectory until a stationary point of the objective function and the other method considered a relaxation scheme to connect multiple stationary points with different trajectories.

Trust-Tech methodology distinguishes itself from existing trajectory methods with following features. Firstly, development of the Trust-Tech methodology is rooted in the theoretical results for characterizing stability regions in autonomous dynamical systems. Secondly, Trust-Tech finds another local optimal solution (or feasible component) from a given local optimal solution (or feasible component) by detecting another stability region in the constructed dynamical system (PGS for computing local optimal solutions and QGS for computing feasible regions). This detection is carried out by detecting the stability boundary between the two stability regions, or more specifically, detecting the decomposition point or the exit point on the stability boundary. Thirdly, the Trust-Tech methodology is able to find multiple disjoint (path-connected) feasible regions and to compute multiple local optimal solutions within each feasible region. Finally, existing local solvers can be effectively incorporated into Trust-Tech methodology for efficiently computing local optimal solutions in the identified stability regions.

3. Trust-Tech-based methodology for solving MINLP

In this section, we present a Trust-Tech-based methodology for solving MINLP problems formulated as follows:

$$\begin{aligned} \min \quad & c(x, y) \\ \text{s.t.} \quad & v_j(x, y) \leq 0 \text{ for } j \in J = \{1, 2, \dots, j_1\}, \\ & h_i(x, y) = 0 \text{ for } i \in I = \{1, 2, \dots, i_1\}, \\ & x^l \leq x \leq x^u, \quad y^l \leq y \leq y^u, \\ & x^l, x, x^u \in R^{n \times 1}, \quad y^l, y, y^u \in Z^{m \times 1}, \end{aligned} \quad (13)$$

where J : the index set for inequality constraints and I : the index set for equality constraints.

In this formulation x is the set of continuous variables that appears either linearly or/and nonlinearly in the formulation, and y represents the set of integer variables in the formulation. The objective function $c(x, y)$, inequality constraints $v_j(x, y)$, and equality constraints $h_i(x, y)$, for the time being, are all continuously differentiable functions of both the continuous variable x_k and the discrete variable y_k , where x_k denotes the k -th element of x and y_k denotes the k -th element of y . If variable y_k can only take value 0 or 1, we call the problem a mixed binary nonlinear programming problem. Otherwise, we will call the problem a mixed integer nonlinear programming problem.

3.1 Existing methods

The above general formulation (13) contains several aspects that are difficult to solve. Features such as nonlinearity in the discrete variables and non-separability of continuous and discrete variables are included. The Generalized Benders Decomposition [23] can be implemented to solve the above mixed-integer nonlinear formulation. In addition, two other algorithms have been proven effective against this problem: the Outer Approximation algorithm [24] and the Equality Relaxation algorithm. These three algorithms make use of projection, outer approximation and relaxation. The basic idea behind all three algorithms is to solve an alternating finite sequence of nonlinear programming (NLP) subproblems, which provide upper bounds for the optimal solution, and mixed-integer linear programming (MILP) master problems, which provide lower bounds for the optimal solution.

3.2 The conceptual algorithm

Compared to solving the continuous optimization problem (2), the major difficulty introduced in solving MINLP problems rests in how to properly handle the integral variables. But the Trust-Tech-based methodology has been developed principally for continuous optimization problems. Hence, the MINLP problem needs first to be converted to a continuous one before Trust-Tech is applied.

On the other hand, as the dimension of the MINLP problem increases, the number of local optimal solutions usually increases as well. Hence, the number of tiers for Trust-Tech to search local optimal solutions in the constructed continuous problem needs to be increased as well. However, if the number of tiers we define as the neighborhood is too large, the entire searching process might be time consuming or unmanageable. This issue needs to be addressed in the proposed Trust-Tech-based methodology. Another issue that needs to be addressed is the development of a method to eliminate the issue of non-convexity arising from the presence of integral variables. Addressing these issues may give our Trust-Tech-based framework advantages for solving continuous constrained problems or easy integration with existing efficient solvers. We hence propose a multi-stage two-phase Trust-Tech-based methodology for solving MINLP problems, which consists of the following basic steps:

- Step 1. Convert the constrained MINLP problem (13) into a continuous and unconstrained problem (15) through relaxation and the penalty function method where integral variables are relaxed into continuous ones;
- Step 2. Construct the dynamical system (16) to search for multiple solutions to the unconstrained problem (15);
- Step 3. Apply a two-stage search in the dynamical system to find a set of local optimal solutions to the original MINLP problem (13). Specifically, for each local optimal solution, a reduced constrained NLP is constructed by determining and fixing the values of integral variables of the MINLP problem. The Trust-Tech-based method is then used to compute a set of local optimal solutions to these reduced NLP problems, from which the optimal solution of the original MINLP problem is determined.

This multi-stage procedure is designed to properly manage the required computational efforts. At each stage, the search procedure of the Trust-Tech-based method is to explore just a small number of tiers around the starting local optimal solution in the search space which we define as the neighborhood. Upon completion of the one-stage search, we determine whether the obtained local optimal

solutions achieve a certain solution quality. If there are no satisfactory local optimal solutions found and sufficient computational time is still available, another stage of the search procedure will continue.

In the meantime, the purpose of a two-phase search procedure within each stage is to eliminate the complicating procedure of determining the optimal values for the integral variables. During the first phase, several local optimal solutions among all the local optimal solutions found are evaluated and used to determine the values of the integral variables. Each valid solution of the integral variables will lead to a corresponding reduced nonlinear programming problem (with fixed values of the integral variables).

In the remaining parts of this section, each step in the Trust-Tech-based method to realize this conceptual algorithm will be described in detail.

3.3 Relaxation and penalty function method

Our proposed Trust-Tech-based methodology for solving the MINLP problem (13) first relaxes the original problem by treating all the discrete variables y_k as continuous variables. The corresponding relaxed problem is then described below:

$$\begin{aligned}
\min \quad & c(x, y) \\
s.t. \quad & v_j(x, y) \leq 0, \text{ for } j \in J = \{1, 2, \dots, j_1\}, \\
& h_i(x, y) = 0, \text{ for } i \in I = \{1, 2, \dots, i_1\}, \\
& x^l \leq x \leq x^u, \quad y^l \leq y \leq y^u, \\
& x^l, x, x^u \in R^{n \times 1}, \quad y^l, y^u \in Z^{m \times 1}, \quad y \in R^{m \times 1}.
\end{aligned} \tag{14}$$

Constraints are then handled by the penalty function method. Penalty terms for different types of constraints are defined as follows.

3.3.1 Inequality constraints and penalty terms

To facilitate our presentation, we define a function g_1 to represent our penalty terms which are defined as follows:

$$\begin{aligned}
g_1(t_{r_1}) &= \begin{cases} 0 & \text{if } t_{r_1} \leq 0, \\ t_{r_1}^2 & \text{if } t_{r_1} > 0, \end{cases} \\
t_{r_1} &= v_j(x, y), \text{ for } r_1 = j \text{ and } 1 \leq j \leq j_1.
\end{aligned}$$

From this construction, the degree of constraint violation is penalized by a quadratic function.

3.3.2 Equality constraints and penalty terms

To facilitate our presentation, we define a function g_2 as follows:

$$\begin{aligned}
g_2(t_{r_2}) &= \begin{cases} 0 & \text{if } t_{r_2} = 0, \\ t_{r_2}^2 & \text{if } t_{r_2} \neq 0, \end{cases} \\
t_{r_2} &= h_i(x, y), \text{ for } r_2 = i + j_1 \text{ and } 1 \leq i \leq i_1.
\end{aligned}$$

From our construction, the degree of constraint violation is again penalized by a quadratic function.

3.3.3 Box constraints for continuous variables

We denote x_k , x_k^u and x_k^l as the k -th element of continuous variables x , where the upper bound is denoted by x^u and lower bound is denoted by x^l , respectively. To facilitate our presentation, we define function g_3 and function g_4 to represent our penalty terms.

Define

$$\begin{aligned}
g_3(t_{r_3}) &= \begin{cases} 0 & \text{if } t_{r_3} \leq 0, \\ t_{r_3}^2 & \text{if } t_{r_3} > 0, \end{cases} \\
g_4(t_{r_4}) &= \begin{cases} 0 & \text{if } t_{r_4} \geq 0, \\ t_{r_4}^2 & \text{if } t_{r_4} < 0, \end{cases} \\
t_{r_3} &= x_k - x_k^u, \text{ for } r_3 = k + j_1 + i_1 \text{ and } 1 \leq k \leq n, \\
t_{r_4} &= x_k - x_k^l, \text{ for } r_4 = k + j_1 + i_1 + n \text{ and } 1 \leq k \leq n.
\end{aligned}$$

The degree of violation of a box constraint is penalized by a quadratic function.

3.3.4 Integral variables and penalty terms

We denote y_k , y_k^u and y_k^l the k -th element of y , where y^u and y^l represent the upper and lower bound, respectively. We define functions g_5 , g_6 and g_7 to facilitate our presentation:

$$\begin{aligned} g_5(t_{r_5}) &= \begin{cases} 0 & \text{if } t_{r_5} \leq 0, \\ 1 + t_{r_5} & \text{if } t_{r_5} > 0, \end{cases} \\ g_6(t_{r_6}) &= \begin{cases} 0 & \text{if } t_{r_6} \geq 0, \\ 1 - t_{r_6} & \text{if } t_{r_6} < 0, \end{cases} \\ g_7(t_{r_7}) &= \begin{cases} \cos(2\pi t_{r_7} - \pi) + 1 & \text{if } y_k^l - \frac{1}{4} \leq t_{r_7} \leq y_k^u + \frac{1}{4}, \\ 0 & \text{else} \end{cases} \\ t_{r_5} &= y_k - y_k^u - 0.25, \quad r_5 = k + j_1 + i_1 + 2n, \\ t_{r_6} &= y_k - y_k^l + 0.25, \quad r_6 = k + j_1 + i_1 + 2n + m, \\ t_{r_7} &= y_k, \quad r_7 = k + j_1 + i_1 + 2n + 2m, \\ 1 &\leq k \leq m. \end{aligned}$$

For each integral variable, depending on its value, only one function among g_5 , g_6 and g_7 is activated. If $y_k^l - 0.25 \leq y_k \leq y_k^u + 0.25$, g_7 is activated to determine the value of the penalty term regarding the integral and box constraint of y_k . If y_k takes integral value, the value of g_7 will be 0 and no penalty is introduced. If $y_k \geq y_k^u + 0.25$, g_5 will be activated and the penalty term will be a linear function of the constraint violation. The situation is similar for $y_k \leq y_k^l - 0.25$ and g_6 .

For a binary integer nonlinear programming problem, we implicitly assume that the problem includes constraints $0 \leq y_k \leq 1$ for any binary variable y_k .

3.4 New unconstrained optimization problem

The new unconstrained optimization problem constructed with relaxation and the penalty function method is given as follows:

$$\min f(x, y), \tag{15}$$

where $x \in R^{n \times 1}$, $y \in R^{m \times 1}$, and

$$\begin{aligned} f(x, y) &= c(x, y) + \alpha \sum_{r_1=1}^{j_1} g_1(t_{r_1}) + \beta \sum_{r_2=j_1+1}^{j_1+i_1} g_2(t_{r_2}) \\ &+ \gamma \sum_{r_3=j_1+i_1+1}^{j_1+i_1+n} g_3(t_{r_3}) + \gamma \sum_{r_4=j_1+i_1+n+1}^{j_1+i_1+2n} g_4(t_{r_4}) \\ &+ \omega \sum_{r_5=j_1+i_1+2n+1}^{j_1+i_1+2n+m} g_5(t_{r_5}) + \omega \sum_{r_6=j_1+i_1+2n+m+1}^{j_1+i_1+2n+2m} g_6(t_{r_6}) \\ &+ \omega \sum_{r_7=j_1+i_1+2n+2m+1}^{j_1+i_1+2n+3m} g_7(t_{r_7}), \end{aligned}$$

where

α : penalty parameter for equality constraints;

β : penalty parameter for inequality constraints;

γ : penalty parameter for a box constraint of continuous variables;

ω : penalty parameter for integral variables.

To search for a set of local optimal solutions to the new unconstrained optimization problem (15), we construct the following gradient dynamical system:

$$\begin{cases} \dot{x} &= -\frac{\partial f(x, y)}{\partial x}, \\ \dot{y} &= -\frac{\partial f(x, y)}{\partial y}. \end{cases} \quad (16)$$

In this gradient system, the procedure of the Trust-Tech-based dynamic decomposition point (DDP) method [28] or the exit point method [30] can then be carried out to search for multiple SEPs, which are local optimal solutions to the unconstrained problem (15).

3.5 Reduced NLP problem

Given a potential local optimal solution (x^*, y^*) , the corresponding reduced NLP problem can be represented as follows:

$$\begin{aligned} \min \quad & c(x, y^*) \\ \text{s.t.} \quad & v_j(x, y^*) \leq 0, \\ & h_i(x, y^*) = 0, \\ & x^l \leq x \leq x^u, \\ & x^l, x, x^u \in R^{n \times 1}. \end{aligned} \quad (17)$$

The reduced NLP problem can be solved by our Trust-Tech-based method for constrained nonlinear programming problems, which is presented in Section 3, or any other existing nonlinear solvers.

3.6 The proposed Trust-Tech-based method for solving MINLP

At this point, we can outline three major steps for our search procedure within one stage as follows:

- Step 1. Locate multiple local optimal solutions of the relaxed MINLP problem using our Trust-Tech-based method.
- Step 2. From each local optimal solution found, determine the value for each integral variable of the given MINLP problem.
- Step 3. Determine the optimal values for continuous variables of the corresponding reduced problem of the original MINLP problem where the optimal values of integral variables have been determined in Step 2.

We describe the basic steps of our proposed search procedure in each stage using the flow chart shown in Fig. 1. The proposed two-stage Trust-Tech-based methodology is presented as follows.

Two-Stage Methodology

Stage 1:

- Step 1.1 Initialization: Choose an initial condition for dynamical system (16).
- Step 1.2 Apply Trust-Tech-based method to locate multiple local optimal solutions starting from the chosen initial condition.
- Step 1.3 Determine the values of the integral variables of the MINLP problem (13) from the local optimal solutions found in Step 1.2.
- Step 1.4 Derive a reduced NLP problem (17) with values of the integral variables fixed at those obtained in Step 1.3.
- Step 1.5 Apply the Trust-Tech-based method to determine a set of local optimal solutions for the values of the reduced NLP problem (17).
- Step 1.6 Determine the optimal solution for the MINLP problem (13) based on the set of local optimal solutions obtained in Step 1.5.

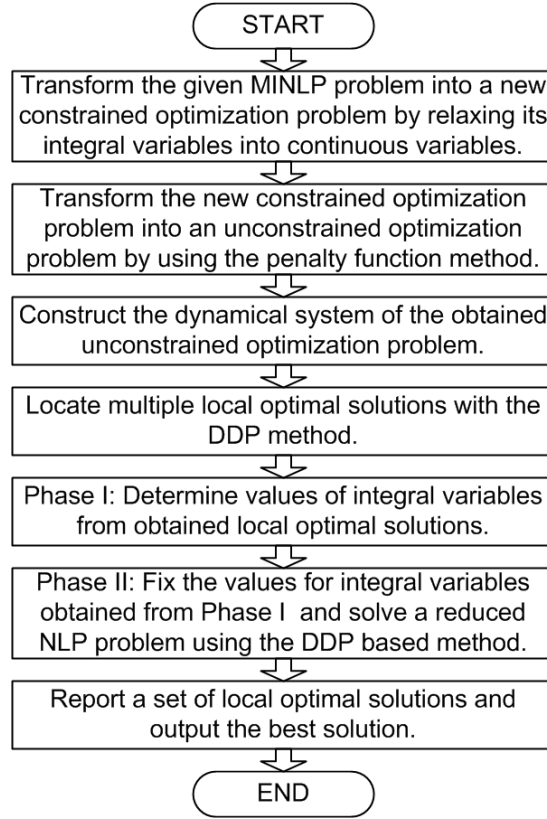


Fig. 1. A flow chart of the Trust-Tech-based methodology for solving MINLP problems in each stage.

Stage 2:

- Step 2.1 Initialization: Use the best local optimal solution identified in Stage 1 as new initial condition for dynamical system (16).
- Step 2.2 Apply the Trust-Tech-based method to locate multiple local optimal solutions starting from the chosen initial condition.
- Step 2.3 Determine the values of the integral variables of MINLP problem (13) from the local optimal solution found in Step 2.2.
- Step 2.4 Derive a reduced NLP problem (17) with values of the integral variables fixed at those obtained in Step 2.3.
- Step 2.5 Apply the Trust-Tech-based method to determine a set of local optimal solutions for the values of the reduced NLP problem (17).
- Step 2.6 Determine the optimal solution for the MINLP problem (13) based on the set of local optimal solutions obtained in Step 2.5.

The first stage applies the Trust-Tech-based method for a breadth-first search (i.e. locate local optimal solutions lying within three to five tiers from the initial local optimal solutions). This stage identifies the top few local optimal solutions from the three-tier to five-tier local optimal solutions and uses the directions from the first local optimal solution to these top local optimal solutions as the search directions to be used in Stage 2.

The second stage applies the Trust-Tech-based method for a depth-first search (i.e. locate multiple local optimal solutions starting from the initial local optimal solution and move along the search directions determined in the first stage).

Generally, the number of local optimal solution increases with the dimension of the optimization problem under study. If the initial condition given to the Trust-Tech-based method is far away

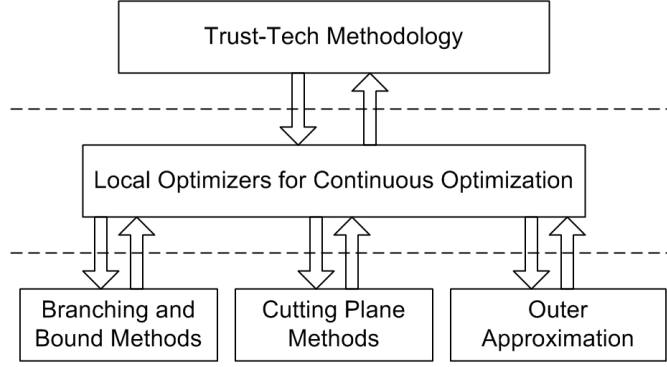


Fig. 2. Architecture of the three-stage Trust-Tech-based methods for solving mixed integer nonlinear programming problems.

from high-quality local optimal solutions, the search procedure needs to go through many tiers to obtain these high-quality local optimal solutions. For large-scale optimization problems, this search procedure can be time consuming or even unmanageable. A naive approach to conquer such difficulty is to reduce the number of search directions. That is, from a located local optimal solution, we only move along a few directions to reach other nearby local optimal solutions. This approach, however, can be effective but it does not make use of any information obtained during the search process.

The issue is then how to determine promising search directions on the basis of all identified local optimal solutions. Since our Trust-Tech method is applied to search a promising region containing high-quality local optimal solutions, we conjecture that nearby local optimal solutions have similar behavior in the sense that they are uniform in terms of their objective values or feasibility of the original MINLP problem. With such a conjecture, high-quality local optimal solutions will cluster together and locate themselves in certain search directions starting from one initial condition. Therefore, if a high-quality local optimal solution is identified in a certain search direction, the search process should continue towards that direction to look for other better solutions. We design a multi-stage procedure of our proposed method to realize such an idea.

During the first stage, the search procedure is focused on exploring a neighborhood of the starting local optimal solution up to certain tiers. The number of tiers should be sufficiently small such that the search process can be completed within a reasonable amount of computational effort. In our simulations, the number of tiers is chosen to be between 3 and 5. We use the first stage as an indication to either increase or decrease the number of tiers for the following stages depending on the computational resources allocated to each stage. Upon the completion of the first stage, a good identified local optimal solution (after rounding integral variables to their nearest values) is used as a new initial condition for the next stage. The process is repeated until a satisfactory local optimal solution is located.

There are several other options available that our Trust-Tech methodology can significantly contribute to solving MINLP problems. For instance, the Trust-Tech methodology obviously can be incorporated with Branch and Bound type algorithm to enhance its pruning scheme. Figure 2 illustrates a framework for such incorporation of the Trust-Tech methodology with existing algorithms.

4. Illustrative examples

In this section, we illustrate our proposed method on several testing problems which appear in the *Handbook of Test Problems in Local and Global Optimization* [40].

4.1 Testing problem 1

The first problem is the following seven-dimensional problem with three continuous and four binary variables:

$$\begin{aligned}
\min \quad & (y_1 - 1)^2 + (y_2 - 2)^2 + (y_3 - 1)^2 + (x_1 - 1)^2 \\
& + (x_2 - 2)^2 + (x_3 - 3)^2 - \ln(y_4 + 1) \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + y_1 + y_2 + y_3 \leq 5, \\
& x_1^2 + x_2^2 + x_3^2 + y_3^2 \leq 5.5, \\
& y_1 + x_1 \leq 1.2, \\
& y_2 + x_2 \leq 1.8, \\
& y_3 + x_3 \leq 2.5, \\
& y_4 + x_1 \leq 1.2, \\
& x_2^2 + y_2^2 \leq 1.64, \\
& x_3^2 + y_3^2 \leq 4.25, \\
& x_3^2 + y_2^2 \leq 4.64, \\
& x_1, x_2, x_3 \geq 0, \\
& y_i \in \{0, 1\} \text{ for } i = 1, \dots, 4.
\end{aligned}$$

4.1.1 Choices of parameters

To determine penalty parameters, we need to determine the range of each variable. From constraints three, four and five, we know $0 \leq x_1 \leq 1.2$, $0 \leq x_2 \leq 1.8$ and $0 \leq x_3 \leq 2.5$ (choose $y_1 = 0$, $y_2 = 0$ and $y_3 = 0$). For each binary variable, we have the box constraint $0 \leq y_i \leq 1$ for each $i = 1, \dots, 4$. The upper bound of the MINLP objective function can be computed as $1 + 4 + 1 + 1 + 4 + 9 + \ln 2 \cong 20.7$ (choose $(x_1, x_2, x_3) = (0, 0, 0)$, and $(y_1, y_2, y_3, y_4) = (0, 0, 0, 1)$). The initial condition of continuous variables (x_1, x_2, x_3) is chosen to be $(0.6, 0.9, 1.25)$, which is located on the center of the range $0 \leq x_1 \leq 1.2$, $0 \leq x_2 \leq 1.8$ and $0 \leq x_3 \leq 2.5$. Again, the initial condition of binary variables is chosen to be $(0.5, 0.5, 0.5)$. In this problem, we allow our search procedure to explore the neighborhood surrounding the first stable equilibrium point up to four tiers. Since this is a small scale problem, we only apply one stage search to identify optimal solutions.

4.1.2 Result

There are 33 stable equilibrium points identified in our simulation for the unconstrained problem (15). The procedure of the Trust-Tech-based DDP method for solving this test system can be illustrated as Fig. 3 in a tier-by-tier manner. In this figure, there are 33 identified decomposition points, which are denoted as x_d^1 to x_d^{33} . It can be seen that four tiers of local optimal solutions have been found with the following:

- Tier-zero (starting local optimal solution): x_s^1 ;
- First-tier local optimal solutions: x_s^2, x_s^3, x_s^4 ;
- Second-tier local optimal solutions: $x_s^5, \dots, x_s^{14}$;
- Third-tier local optimal solutions: $x_s^{15}, \dots, x_s^{27}$;
- Fourth-tier local optimal solutions: $x_s^{28}, \dots, x_s^{34}$.

After rounding all the binary variables to their nearest integral values, we sort all these rounded stable equilibrium points in non-decreasing order according to their values in the MINLP objective function. The thirteenth rounded stable equilibrium point $(0, 0.8183, 1.9127, 1, 1, 0, 2)$ (obtained from rounding the binary variables of the located stable equilibrium point $(0, 0.8183, 1.9127, 1, 0.9977, 0.0024, 1.7089)$ to their nearest integral values) possesses the smallest value of the MINLP objective function. However, we know that all y_i variables should possess binary values. This rounded stable equilibrium point may possess the smallest value of the MINLP due to its strong violation of some constraints. Therefore, we skip to the second best candidate, the fourth rounded stable equilibrium point $(0.2195, 0.8183, 1.9127, 1, 1, 0, 1)$ (obtained from rounding the binary variables of the located stable equilibrium point $(0.2195, 0.8183, 1.9127, 0.9991, 0.9977, 0.0024, 0.9996)$ to their nearest integral values). From this rounded stable equilibrium point, we determine the values of the binary variables to be $(1, 1, 0, 1)$ and construct the following reduced NLP problem:

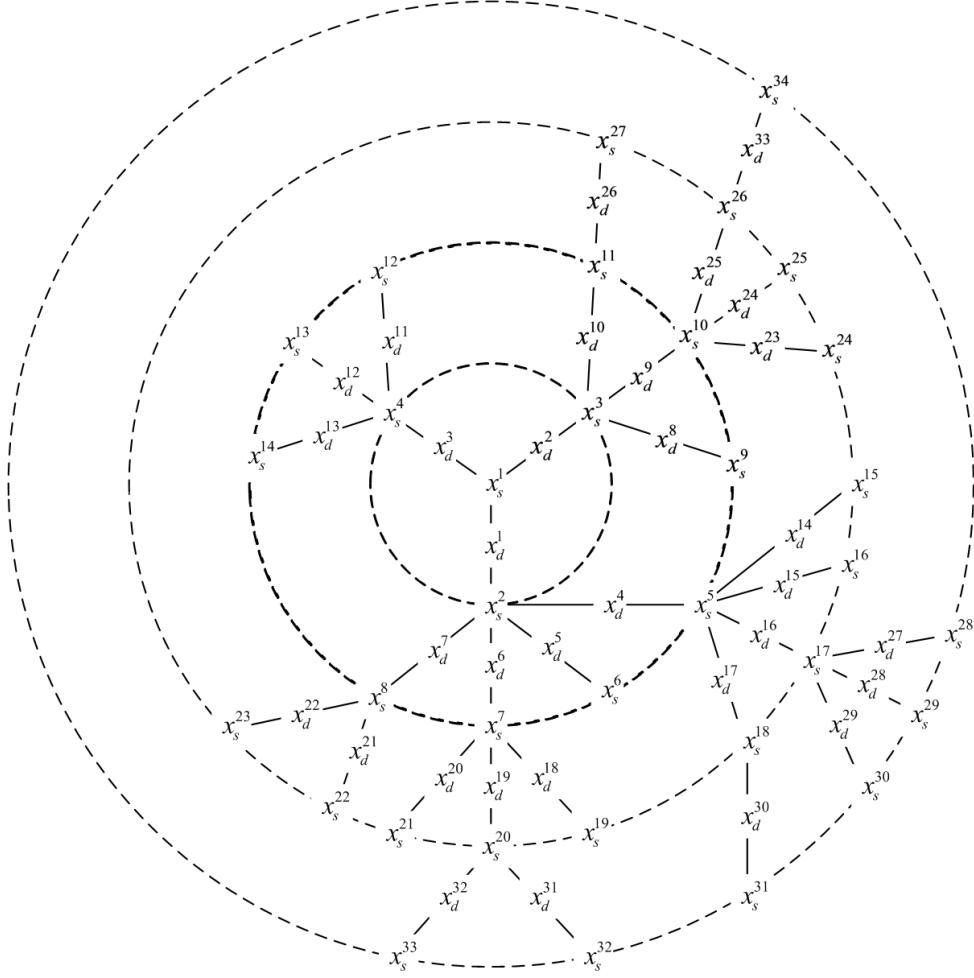


Fig. 3. Result of stage I of problem 1. There are 34 stable equilibrium points within the fourth tier of the starting SEP x_s^1 . The global optimal solution of this stage, x_s^4 , is located on the first tier of the starting SEP.

$$\begin{aligned}
& \min \quad (x_1 - 1)^2 + (x_2 - 2)^2 + (x_3 - 3)^2 + 2 - \ln 2 \\
& s.t. \quad x_1 + x_2 + x_3 \leq 3, \\
& \quad \quad x_1^2 + x_2^2 + x_3^2 \leq 5.5, \\
& \quad \quad x_1 \leq 0.2, \\
& \quad \quad x_2 \leq 0.8, \\
& \quad \quad x_3 \leq 2.5, \\
& \quad \quad x_2^2 \leq 0.64, \\
& \quad \quad x_3^2 \leq 4.25, \\
& \quad \quad x_3^2 \leq 3.64.
\end{aligned} \tag{18}$$

We may further reduce this NLP problem through the following observations:

- (1) We may drop constraint $x_3^2 \leq 4.25$ since $x_3^2 \leq 3.64$ implies that $x_3^2 \leq 4.25$.
- (2) $x_3^2 \leq 3.64$ can be re-written as constraint $-1.908 \leq x_3 \leq 1.908$. Therefore, we may drop constraint $x_3 \leq 2.5$.
- (3) $x_2^2 \leq 0.64$ can be re-written as constraint $-0.8 \leq x_2 \leq 0.8$. Therefore, we drop the constraint $x_2 \leq 0.8$.
- (4) $x_1 \leq 0.2$, $-1.908 \leq x_3 \leq 1.908$ and $-0.8 \leq x_2 \leq 0.8$ directly imply that $x_1 + x_2 + x_3 \leq 3$ and $x_1^2 + x_2^2 + x_3^2 \leq 5.5$.

Our reduced NLP problem is now equivalent to the following optimization problem:

$$\begin{aligned}
\min \quad & (x_1 - 1)^2 + (x_2 - 2)^2 + (x_3 - 3)^2 + 2 - \ln 2 \\
s.t. \quad & x_1 \leq 0.2, \\
& -0.8 \leq x_2 \leq 0.8, \\
& -1.908 \leq x_3 \leq 1.908.
\end{aligned} \tag{19}$$

To minimize the value of objective function, we only need to choose (x_1, x_2, x_3) within the constraints such that its Euclidean distance to the point $(1, 2, 3)$ is minimized. Clearly, $(0.2, 0.8, 1.908)$ will be the optimal solution of our reduced NLP problem. After combining this value with that of the binary variables, we will obtain the optimal solution of the MINLP problem as shown in Table I.

Table I. The best solution for testing problem 1.

x_1	x_2	x_3	y_1	y_2	y_3	y_4	$c(x, y)$
0.2	0.8	1.908	1	1	0	1	4.5796

In this testing problem, for the purpose of illustration, we solve the reduced NLP problem through careful observation. In general, solving reduced the NLP problem will require other existing continuous nonlinear optimization algorithms or our DDP method. In the following testing problems 2 and 3, the reduced NLP problem will be solved using the DDP method.

4.2 Testing problem 2

The second problem is the following five-dimensional MINLP problem with two continuous and three binary variables:

$$\begin{aligned}
\min \quad & 2x_1 + 3x_2 + 1.5y_1 + 2y_2 - 0.5y_3 \\
s.t. \quad & x_1^2 + y_1 = 1.25, \\
& x_2^{1.5} + 1.5y_2 = 3, \\
& x_1 + y_1 \leq 1.6, \\
& 1.333x_2 + y_2 \leq 3, \\
& -y_1 - y_2 + y_3 \leq 0, \\
& x_1, x_2 \geq 0, \\
& y_i \in \{0, 1\} \text{ for } i = 1, 2, 3.
\end{aligned} \tag{20}$$

We next present the results after applying Stage 1 and Stage 2 of the Trust-Tech-based method. Five local optimal solutions are found for the unconstrained problem in Stage 1. Based on these potential local optimal solutions, five reduced problems are constructed and there are a total of eight local optimal solutions found after the application of Stage 2. Of these eight solutions, the best solution is the one shown in Table II whose objective function value is 7.6672.

Table II. The best solution for testing problem 2.

x_1	x_2	y_1	y_2	y_3	$c(x, y)$
1.1180	1.3104	0	1	1	7.6672

The procedures of Stage 1 for finding multiple local optimal solutions to the unconstrained problem and those of Stage 2 for finding a set of local optimal solutions to the reduced problems are illustrated in Figs. 4(a) and 4(b), respectively.

4.3 Testing problem 3

The third test problem is an eleven-dimensional MINLP problem with three continuous and eight binary variables defined as follows:

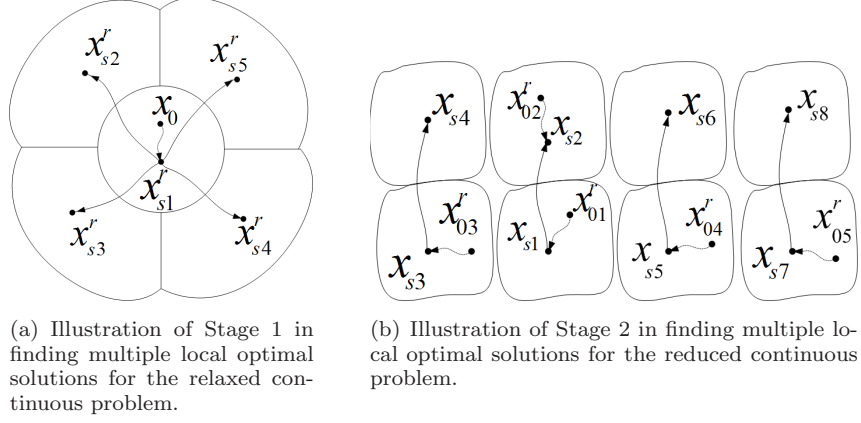


Fig. 4. Illustration of the solution to test problem 2.

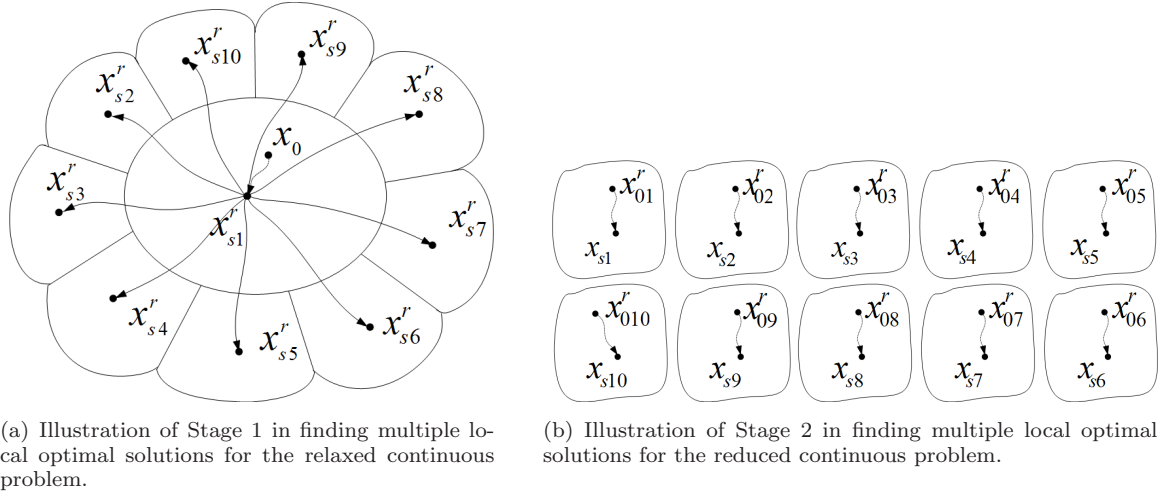


Fig. 5. Result of test problem 3.

$$\begin{aligned}
\min \quad & -x_1 x_2 x_3 \\
\text{s.t.} \quad & x_1 + 0.1^{y_1} 0.2^{y_2} 0.15^{y_3} = 1, \\
& x_2 + 0.05^{y_4} 0.2^{y_5} 0.15^{y_6} = 1, \\
& x_3 + 0.02^{y_7} 0.06^{y_8} = 1, \\
& -y_1 - y_2 - y_3 \leq -1, \\
& -y_4 - y_5 - y_6 \leq -1, \\
& -y_7 - y_8 \leq -1, \\
& 3y_1 + y_2 + 2y_3 + 3y_4 + 2y_5 + y_6 + 3y_7 + 3y_8 \leq 10, \\
& 0 \leq x_1, x_2, x_3 \leq 1, \\
& y_i \in \{0, 1\} \text{ for } i = 1, \dots, 8.
\end{aligned} \tag{21}$$

We next present the results after applying Stage 1 and Stage 2 of the Trust-Tech-based methodology. There are ten local optimal solutions found after the application of Stage 2. Of the ten solutions, the best optimal solution is the one presented in Table III whose objective function value is -0.0249.

Table III. The best solution for testing problem 3.

x_1	x_2	x_3	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	$c(x, y)$
1.000	0.8442	0.1732	0	0	0	0	1	0	0	0	-0.1462

The process of Stage 1 and Stage 2 of the Trust-Tech based method in finding multiple local optimal solutions for the unconstrained problem and reduced problems is illustrated in Figs. 5(a) and 5(b), respectively.

5. Concluding remarks

Many smart grid applications can be formulated as mixed integer nonlinear programs. In this paper, we have developed a two-stage Trust-Tech-based methodology to systematically compute all the local optimal solutions for general MINLP problems. The methodology consists of two distinct stages: Stage I, for a given MINLP problem, constructs a new continuous unconstrained problem through relaxation and the penalty function method, and it searches for all the local optimal solutions of the unconstrained problem. Stage II defines a reduced constrained problem for each local optimal solution found in Stage I and computes all the local optimal solutions for the reduced problem, from which the optimal solution of the original MINLP problem is determined.

From a computational viewpoint, for general constraint MINLP problems, it may well be due to the lack of computational methods for computing the complete set of decomposition points that the Trust-Tech-based methodology can only compute multiple local optimal solutions for the associated unconstrained NLP problem and the corresponding reduced constrained NLP problems. In other words, the Trust-Tech-based methodology can only compute multiple local optimal solutions, instead of all the local optimal solutions for general MINLP problems.

The Trust-Tech-based methodology is valuable and can be of significant importance in smart grid related applications involving MINLP problems. Such importance can be justified in the following sense: since a sophisticated local-type method can only provide local optimal solutions in a restricted local region while a global-type algorithm can only provide sparsely approximated solutions within reasonable computational effort, the Trust-Tech-based methodology represents an enabling technology for:

- 1) Enhancing the functionality of local-type methods to effectively find optimal solutions in a broader region;
- 2) Enhancing the functionality of global-type methods to effectively find near-global optimal solutions;
- 3) Providing more flexibility in integrating local-type methods and global-type algorithms to effectively find the global optimal solution.

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